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Werk

Titel: Inventiones Mathematicae

Verlag: Springer

Jahr: 1987

Kollektion: Mathematica

Werk Id: PPN356556735_0090

PURL: http://resolver.sub.uni-goettingen.de/purl?PID=PPN356556735_0090|LOG_0012

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Invent. math. 90, 153-167

Inventiones mathematicae
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Rational intersection cohomology of quotient varieties. II

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Given a linear action of a complex reductive group G on a complex projective variety X one can define a projective "quotient" variety $X//G$ using Mumford's geometric invariant theory [18]. From the construction of this quotient it is not immediately obvious how its topology is related to the topology of X and the group action. However, when X is nonsingular it is possible (at least if one technical condition is satisfied) to compute the intersection Betti numbers of $X//G$ in terms of the rational cohomology of certain nonsingular linear sections Z of X and the classifying spaces of certain reductive subgroups H of G (see [15]). The technical condition is that there should exist at least one point of X which is (properly) stable for the action of G: this is satisfied in most interesting examples such as those coming from moduli problems. The sections of X and subgroups of G which occur have a simple description in terms of the representation of G which induces the linear action on X.

It thus seems reasonable to ask what happens when X is singular. The aim of this paper is to show that the procedure for computing the intersection Betti numbers of $X//G$ can be generalised to the singular case (see (2.25) and (2.28)). However, now one needs to know not only the intersection cohomology of the various sections Z of X but also how they are positioned relative to the singularities of X. In the special case when they meet the singularities transversely then one has exactly the same formulas as when X is nonsingular except that ordinary cohomology is replaced by intersection cohomology throughout.

The situation which it is most important to understand is that in which every semistable point is stable, and this case is considered in detail in § 2 after the necessary preliminaries have been covered in § 1. Some brief remarks are made about the more general case when one only assumes that there exist some stable points. A simple example is worked out (2.27), which shows that the formulas can also be used in reverse to compute the intersection Betti numbers of X from knowledge of the intersection Betti numbers of the quotient $X//G$ (even simply from the knowledge that these vanish in dimensions greater than $\dim_{\mathbb{R}} X//G$).

Acknowledgement. I would like to thank the I.H.E.S. for its hospitality while this paper was written.

§1. This section describes very briefly the necessary background to §2. For more details see [13, 15] and [18].

Throughout this paper $IH^*(V)$ will denote the rational intersection cohomology with respect to the middle perversity of a variety V (as defined in [4, 5, 9]). $H^*(V)$ will denote ordinary cohomology with rational coefficients. When V is nonsingular or has only finite quotient singularities then $IH^*(V)=H^*(V)$. The universal classifying bundle for a compact group K is denoted by $EK \to BK$. If K acts on V then $H_K^*(V) =_{\mathrm{df}} H^*(V \times_K EK)$ is the rational equivariant cohomology of V. The rational equivariant intersection cohomology $IH_K^*(V)$ of V can be defined in a similar way by interpreting $IH^*(V \times_K EK)$ appropriately (see [3] and also [12]). Let

$$P_t(V) = \sum_{i \geqq 0} t^i \dim H^i(V),$$

$$IP_t(V) = \sum_{i \geqq 0} t^i \dim IH^i(V)$$

and

$$IP_t^K(V) = \sum_{i \geqq 0} t^i \dim IH_K^i(V)$$

be the Poincaré series, intersection Poincaré series and equivariant intersection Poincaré series of V.

Let X be a closed subvariety of a complex projective space $\mathbb{P}_n$, and let G be a complex reductive subgroup of $GL(n+1;\mathbb{C})$ such that X is invariant under the linear action of G on $\mathbb{P}_n$. Since G is reductive it is the complexification of a maximal compact subgroup K. It will be assumed for simplicity that G is connected, but this is not an essential restriction (cf. [15], §1).

The geometric invariant theory quotient $X//G$ of X by G is the projective variety whose homogeneous coordinate ring consists of the G-invariant polynomials on X. There is a G-invariant surjective morphism $\phi\colon X^{ss} \to X//G$ where X^{ss} is the set of semistable points for the action. A point of X is semistable if and only if it is semistable for the action of G on the ambient projective space $\mathbb{P}_n$.

When X^{ss} coincides with its open subset X^s consisting of the (properly) stable points then ϕ induces a homeomorphism $X^{ss}/G \to X//G$, and the stabiliser in G of every semistable point is finite. Therefore if in addition X is nonsingular $X//G$ has only finite quotient singularities. This case is studied in [13], where a formula is derived for the Betti numbers of $X//G$. In fact in all cases when X is nonsingular this formula gives one the equivariant Betti numbers $\dim H_K^i(X^{ss})$ of X^{ss}; in the special case when $X^{ss}=X^s$ these are the same as the Betti numbers of $X//G$. The formula comes from the existence of a certain stratification $\{S_\beta | \beta \in \mathscr{B}\}$ of X with the following properties.

(1.1) (a) S_β is a locally closed G-invariant nonsingular subvariety of X.

(b) X^{ss} coincides with the unique open stratum S_0.

(c) The stratification is equivariantly perfect, i.e.

$$P_t^K(X) = P_t^K(X^{ss}) + \sum_{\beta \neq 0} t^{2\lambda(\beta)} P_t^K(S_\beta)$$

where $\lambda(\beta)$ is the (complex) codimension of S_β in X. (The strata S_β may not be connected and their components may be of different dimensions: thus we should really consider the components separately here. However to avoid cumbersome notation we shall ignore this possibility.)

(d) If $\beta \neq 0$ then there is a proper nonsingular subvariety Z_β of X acted on by a reductive subgroup Stab β of G such that

$$P_t^K(S_\beta) = P_t^{K(\beta)}(Z_\beta^{ss})$$

where $K(\beta) = K \cap \text{Stab}\,\beta$ and Z_β^{ss} is the set of semistable points of Z_β with respect to an appropriate linearisation of the action of Stab β. Moreover Z_β is the intersection of X with a linear subspace of $\mathbb{P}_n$ which is determined by the representation of K on $\mathbb{C}^{n+1}$.

Since $P_t^K(X) = P_t(X)\,P_t(BK)$ if K is connected, this gives the following inductive formula for the equivariant Betti numbers of X^{ss} when X is nonsingular.

$$P_t^K(X^{ss}) = P_t(X)\,P_t(BK) - \sum_{\beta \neq 0} t^{2\lambda(\beta)}\,P_t^{K(\beta)}(Z_\beta^{ss}). \tag{1.2}$$

From this an explicit formula can be derived for $P_t^K(X^{ss})$ and hence for the Betti numbers of $X//G$ when $X^{ss} = X^s$ (see [13], Thm. 8.12). It is this formula which will be generalised in §2 to the case when X is singular.

When X is nonsingular the stratification $\{S_\beta | \beta \in \mathscr{B}\}$ associated to the action of G on X can be obtained from the stratification of the ambient projective space $\mathbb{P}_n$ by simply intersecting each stratum with X. We can define a stratification $\{S_\beta | \beta \in \mathscr{B}\}$ of X when X is singular in the same way.

The indexing set $\mathscr{B}$ can be described as follows. Let T be a maximal torus of G and let $\alpha_0, \ldots, \alpha_n$ be the weights of the representation of G which defines its linear action on X. Let W be the Weyl group of G and let t be the Lie algebra of the maximal compact subgroup of T which we may assume is contained in K. Using some fixed W-invariant inner product on t we may identify t with its dual. Thus we may think of the weights $\alpha_0, \ldots, \alpha_n$ as integral points of t. Given any nonempty set I of these weights, its convex hull Conv I in t contains a unique point β which is closest to 0 with respect to the fixed inner product. We define $\mathscr{B}$ to be the set of all such points β which lie in some chosen positive Weyl chamber t_+ in t.

We may suppose that T acts diagonally on $\mathbb{P}_n$ via

$$t \to \text{diag}(\alpha_0(t), \ldots, \alpha_n(t)).$$

For each $\beta \in \mathscr{B}$ let Z_β be the intersection of X with the linear subspace of $\mathbb{P}_n$ defined by the vanishing of those coordinates x_i such that $\alpha_i \cdot \beta \neq \|\beta\|^2$. Then Z_β is a union of connected components of the fixed point set of the action on X of the subtorus T_β of T generated by β. Let Y_β be the intersection of X with the subset

$$\{(x_0; \ldots; x_n) \in \mathbb{P}_n | x_i = 0 \text{ if } \alpha_i \cdot \beta < \|\beta\|^2 \text{ and } x_j \neq 0$$
$$\text{for some } j \text{ such that } \alpha_j \cdot \beta = \|\beta\|^2\}$$

of $\mathbb{P}_n$. Equivalently

$$Y_\beta = \{x \in X \mid \lim_{t \to \infty} \exp(-it\beta)x \in Z_\beta\}. \tag{1.3}$$

There is an obvious projection p_β: $Y_\beta \to Z_\beta$.

Let Stab β be the stabiliser of β under the adjoint action of G on its Lie algebra. Then Stab β acts linearly on Z_β. We may modify the linearisation of this action without changing the action itself by multiplying it by any character of Stab β. It is proved in [13] that when X is nonsingular

$$S_\beta = GY_\beta^{ss}$$

where $Y_\beta^{ss} = p_\beta^{-1}(Z_\beta^{ss})$ and Z_β^{ss} is the set of semistable points in Z_β under the action of Stab β with an appropriately modified linearisation (see [13], 8.11). (In §2 we shall consider a different linearisation for which no point of Z_β is semistable, but Z_β^{ss} will still be the unique open stratum in the stratification of Z_β with respect to this new linearisation.) If $g \in G$ and $y \in Y_\beta^{ss}$ then $gy \in Y_\beta^{ss}$ if and only if g lies in the parabolic subgroup $P_\beta = B$ Stab β of G, where B is the Borel subgroup associated to t_+. In fact

$$S_\beta \cong G \times_{P_\beta} Y_\beta^{ss}. \tag{1.4}$$

The same is true when X is singular simply because it is true for the ambient projective space $\mathbb{P}_n$.

§2. Now let us consider how to generalise formula (1.2) to the case when X may be singular. The result we are aiming for is Theorem (2.25).

Let $\{S_\beta \mid \beta \in \mathscr{B}\}$ be the stratification of X associated to the action of G as described in §1. There is a partial ordering on $\mathscr{B}$, given by $\beta < \gamma$ if and only if $\|\beta\| < \|\gamma\|$, which satisfies

$$\bar{S}_\beta \subseteq \bigcup_{\gamma \geqq \beta} S_\gamma$$

for every $\beta \in \mathscr{B}$. (This is true for $\mathbb{P}_n$ by [13], 12.16 and hence also for any $X \subseteq \mathbb{P}_n$). It follows that we can write

$$\mathscr{B} = \{\beta_0 = 0, \beta_1, \beta_2, \ldots, \beta_s\}$$

in such a way that for $0 \leqq j \leqq s$

$$U_j = S_{\beta_0} \cup S_{\beta_1} \cup \ldots \cup S_{\beta_j}$$

is an open subset of X and S_{β_j} is closed in U_j. Therefore there exist long exact sequences of equivariant intersection cohomology

$$\ldots \to IH_K^i(U_j, U_{j-1}) \to IH_K^i(U_j) \to IH_K^i(U_{j-1}) \to \ldots \tag{2.1}$$

for each j (see [8], 1.3).

(2.2) **Proposition.** *These long exact sequences split into short exact sequences*

$$0 \to IH_K^i(U_j, U_{j-1}) \to IH_K^i(U_j) \to IH_K^i(U_{j-1}) \to 0$$

for each i and j. Thus

$$\dim IH^i_K(X^{ss}) = \dim IH^i_K(X) - \sum_{1 \leqq j \leqq s} \dim IH^i_K(U_j, U_{j-1})$$

for each i.

Proof (cf. [16], 2.6). It is enough to prove that each restriction map

$$IH^i_K(U_j) \to IH^i_K(U_{j-1})$$

is surjective.

By Hironaka's equivariant resolution of singularities theorem, announced in [11], 9, Remark 8 (see also [11], Bibliography with comments, 24) there is a G-equivariant desingularisation $\pi\colon Y \to X$ of X obtained by blowing up X along a sequence of G-invariant nonsingular closed subvarieties. Then

$$\pi\colon \pi^{-1}(U_j) \to U_j$$

is a desingularisation of the open subset U_j of X for each j. Hence by the decomposition theorem of Beilinson, Bernstein, Deligne and Gabber ([2], 6.2.5; see also [6], §2 and [17], §12) $IH^*(U_j)$ is a direct summand of $H^*(\pi^{-1}(U_j))$. Thus there exist projections from $H^*(\pi^{-1}(U_j))$ to $IH^*(U_j)$ for each j, and we may assume that these fit into a commutative diagram

$$\begin{array}{ccc} H^*(\pi^{-1}(U_j)) & \longrightarrow & H^*(\pi^{-1}(U_{j-1})) \\ \downarrow & & \downarrow \\ IH^*(U_j) & \longrightarrow & IH^*(U_{j-1}). \end{array}$$

The same is true for equivariant cohomology and equivariant intersection cohomology (see the discussion preceding [16], 2.13). Thus it suffices to prove that the restriction maps

$$H^*_K(\pi^{-1}(U_j)) \to H^*_K(\pi^{-1}(U_{j-1}))$$

are surjective for all j.

Let us consider the blow-up $\rho\colon W \to X$ of X along any closed G-invariant subvariety V. Let E be the exceptional divisor. If L is the hyperplane line bundle on X then $L_d = \rho^* L^{\otimes d} \otimes \mathcal{O}(-E)$ is a very ample line bundle on W for d sufficiently large. Moreover the action of G on X lifts to an action on W which is linear with respect to L_d for any d. It is shown in [14], §3 that when X is nonsingular the semistable stratum W^{ss} of W is independent of d, provided that d is large enough, and that $\rho^{-1}(W^{ss})$ is a union of strata in the stratification of W induced by the linear G-action (see [14], 3.4). Exactly the same argument shows that if d is large enough the stratification of W is independent of d and $\rho^{-1}(S_\beta)$ is a union of strata in this stratification for each $\beta \in \mathscr{B}$. (One first shows that this is true for the case when G is a torus and then deduces it in general from [14], 3.3.) It follows immediately from considering the ambient projective space that the same is true when X is singular.

Therefore there exists a linear action of G on the desingularisation Y of X which lifts the action of G on X and is such that, for every $\beta \in \mathscr{B}$, the inverse

image $\pi^{-1}(S_\beta)$ of S_β is a union of strata in the stratification $\{\Sigma_\gamma | \gamma \in \Gamma\}$ of Y associated to this linear action. Since each U_j is a union of strata S_β of X, it follows that $\pi^{-1}(U_j)$ is a union of strata Σ_γ of Y for each j.

Since Y is a nonsingular projective variety we know from [13], 5.4 that the restriction map

$$H_K^*(V) \to H_K^*(V - \Sigma_\gamma)$$

is surjective for each stratum Σ_γ of Y and each open subset V of Y which is a union of strata and contains Σ_γ as a closed subset. Therefore the same is true when Σ_γ is replaced by any union V' of strata such that V' is a closed subset of V. In particular it follows that the restriction map

$$H_K^*(\pi^{-1}(U_j)) \to H_K^*(\pi^{-1}(U_{j-1}))$$

is surjective for each j. This completes the proof of the proposition.

(2.3) *Remark.* The stratification $\{S_\beta | \beta \in \mathscr{B}\}$ can be regarded as the Morse stratification of a real valued function f on X which is the norm-square of the moment map (see [13]), and 2.2 can be regarded as expressing the Morse inequalities (in fact in this case equalities) for f on X with respect to equivariant intersection homology. Goresky and MacPherson have studied Morse theory for singular varieties and intersection homology in [7]: however f is not in general a Morse function in their sense.

Next note that because X is a projective variety $IH_K^*(X)$ has a special form. Indeed by [3], 4.2.2 we have

$$IH_K^*(X) \cong IH^*(X) \otimes H^*(BK) \tag{2.4}$$

where BK is the classifying space of K.

We also have

(2.5) **Lemma.** *If every semistable point of X is stable then*

$$IH_K^*(X^{ss}) \cong IH^*(X//G).$$

Proof. It is shown in [13] that when $X^{ss} = X^s$ and X is nonsingular X^{ss} retracts onto a certain K-invariant compact submanifold $\mu^{-1}(0)$ of X such that $\mu^{-1}(0)/K$ is homeomorphic to $X//G$. In fact

$$X^{ss} = G\mu^{-1}(0) \cong G \times_K \mu^{-1}(0) \tag{2.6}$$

by [13], 7.4. It is easy to see that the same is true when X is singular by considering the ambient projective space $\mathbb{P}_n$. Since intersection homology is a homeomorphism invariant ([9], 4.1) it follows that

$$IH^*(X//G) \cong IH^*(\mu^{-1}(0)/K).$$

Moreover since the coset space G/K is homeomorphic to Euclidean space, 2.6 implies that the inclusion of $\mu^{-1}(0)$ in X^{ss} is normally nonsingular ([9], 5.4.1) and that

$$IH_K^*(X^{ss}) \cong IH_K^*(\mu^{-1}(0)).$$

Finally since $X^{ss}=X^s$ the stabiliser in G of every $x\in X^{ss}$ is finite, and hence in particular the stabiliser in K of every $x\in\mu^{-1}(0)$ is finite. This implies that

$$IH_K^*(\mu^{-1}(0))\cong IH^*(\mu^{-1}(0)/K)$$

because the classifying space of any finite group has trivial rational cohomology (cf. [13], 5.6). The result follows.

From (2.2), (2.3) and (2.5) we immediately obtain the formula

$$IP_t(X//G)=IP_t(X)\,P_t(BK)-\sum_{1\leqq j\leqq s} IP_t^K(U_j,U_{j-1}). \tag{2.7}$$

Now consider the relative intersection cohomology groups $IH_K^*(U_j,U_{j-1})$. For each j we have $U_{j-1}=U_j-S_\beta$ where $\beta=\beta_j$, and S_β is closed in U_j which is an open subset of X. Thus by excision (see [8], 1.5).

$$IH_K^i(U_j,U_{j-1})=(IH_K^i(U_\beta,U_\beta-S_\beta) \tag{2.8}$$

for any open neighbourhood U_β of S_β in U_j. By 1.4 we have

$$S_\beta\cong G\times_{P_\beta} Y_\beta^{ss} \tag{2.9}$$

where P_β is the product of the Borel subgroup B of G and the reductive subgroup Stab β. Since $G=KB$ it follows that

$$S_\beta=KY_\beta^{ss}=K\times_{K(\beta)} Y_\beta^{ss} \tag{2.10}$$

where $K(\beta)=K\cap\mathrm{Stab}\,\beta$. Therefore by the slice theorem for compact Lie groups (see e.g. [10], Prop. 27.2) applied to the ambient projective space, there is an open neighbourhood of Y_β^{ss} in U_j of the form $U_\beta=KV_\beta$ where V_β is a $(K,K(\beta))$-slice in X near Y_β^{ss}. That is, V_β is the intersection of X with a $K(\beta)$-invariant neighbourhood of Y_β^{ss} in a submanifold of the ambient projective space and there is a homeomorphism

$$U_\beta=KV_\beta\cong K\times_{K(\beta)} V_\beta \tag{2.11}$$

induced by the natural map $K\times V_\beta\to KV_\beta$.

Recall that intersection cohomology is a homeomorphism invariant ([9], 4.1). Therefore (2.8), (2.10), and (2.11) imply that

$$\begin{aligned} IH_K^*(U_j,U_{j-1})&\cong IH_K^*(K\times_{K(\beta)} V_\beta, K\times_{K(\beta)}(V_\beta-Y_\beta^{ss}))\\ &\cong IH_{K(\beta)}^*(V_\beta,V_\beta-Y_\beta^{ss}) \end{aligned} \tag{2.12}$$

(see [1], § 13). Moreover, since the open neighbourhood $U_\beta=KV_\beta$ of V_β in X is homeomorphic to $K\times_{K(\beta)} V_\beta$ the inclusion of V_β in U_β is a normally nonsingular inclusion ([9], 5.4.1). Therefore it follows from [9], 1.11 and 5.4.1 that

$$IH_{K(\beta)}^i(V_\beta,V_\beta-Y_\beta^{ss})\cong IH_{K(\beta)}^{i-k(\beta)}(U_\beta,U_\beta-Y_\beta^{ss}) \tag{2.13}$$

where $k(\beta)$ is the dimension of the coset space $K/K(\beta)$. Using excision ([8], 1.5) we deduce the following result.

(2.14) **Proposition.** *When $X^{ss}=X^s$ the intersection Betti numbers of $X//G$ are given by the formula*

$$IP_t(X//G)=IP_t^K(X^{ss})=IP_t(X)\,P_t(BK)-\sum_{0\neq\beta\in\mathscr{B}} t^{k(\beta)}\,IP_t^{K(\beta)}(U_\beta, U_\beta-Y_\beta^{ss})$$

where U_β is any $K(\beta)$-invariant neighbourhood of Y_β^{ss} in X such that Y_β^{ss} is closed in U_β.

(2.15) *Remark.* By Poincaré duality ([9], 5.3) it is only necessary to compute $IP_t(X//G)$ modulo t^{d+1} where $d=\dim X//G$. In particular if one can show that $IH^{i-k(\beta)}_{K(\beta)}(U_\beta, U_\beta-Y_\beta^{ss})=0$ for $i\leqq d$ whenever $0\neq\beta\in\mathscr{B}$, and if $X^{ss}=X^s$, then by (2.14)

$$IP_t(X//G)=IP_t^K(X^{ss})\cong IP_t(X)\,P_t(BK) \bmod t^{d+1}.$$

(2.16) *Remark.* The assumption that $X^{ss}=X^s$ was only needed to show that $IP_t(X//G)=IP_t^K(X^{ss})$. The formula for $IP_t^K(X^{ss})$ given in (2.14) is valid even when $X^{ss}\neq X^s$.

Now let us consider how we might compute each relative intersection cohomology group $IH^i_{K(\beta)}(U_\beta, U_\beta-Y_\beta^{ss})$ in terms of $IH^*(W_\beta, W_\beta-Y_\beta)$ where W_β is an open neighbourhood of Y_β in X_β such that Y_β is closed in W_β. Our aim is to show that there is a stratification $\{\Sigma_\gamma|\gamma\in\mathscr{C}\}$ of X such that U_β can be chosen to be the complement in W_β of some of the strata Σ_γ (see 2.21 below). This stratification will be induced by a suitable linear reductive group action on X in just the same way as the original stratification $\{S_\beta|\beta\in\mathscr{B}\}$ was obtained from the given linear action of G on X. It will then be possible to mimic the proof of (2.14) to show that there is an inductive formula

$$IP_t^{K(\beta)}(U_\beta, U_\beta-Y_\beta^{ss})=IP_t(W_\beta, W_\beta-Y_\beta)\,P_t(BK(\beta))-\sum_\gamma t^{k(\beta,\gamma)}\,IP_t^{K(\beta,\gamma)}(U_{\beta,\gamma}, U_{\beta,\gamma}-Y_{\beta,\gamma}^{ss})$$

where the terms $IP_t^{K(\beta,\gamma)}(U_{\beta,\gamma}, U_{\beta,\gamma}-Y_{\beta,\gamma}^{ss})$ are of exactly the same form as $IP_t^{K(\beta)}(U_\beta, U_\beta-Y_\beta^{ss})$ itself but defined using the new group action (see (2.25)(b) below). The argument is based on the fact that, if S and Y are closed subvarieties of a quasi-projective variety U, then the relative intersection cohomology of the pairs $(U, U-S)$ and $(U-Y, U-Y-S)$ is

$$IH^*(U,(U-Y)\cup(U-S))=IH^*(U, U-(YS))$$

(see [8], 1.3 and 1.5).

If m is a suitable arbitrarily large positive integer then $m\beta$ is an integral point of $t^*\cong t$ (provided that the fixed inner product is chosen to be rational) and there is a character $\operatorname{Stab}\beta\to\mathbb{C}^*$ of $\operatorname{Stab}\beta$ whose derivative is $m\beta$ regarded as an element of $(\operatorname{Lie} K(\beta))^*\cong\operatorname{Lie} K(\beta)$. We may modify the linearisation of the action of $\operatorname{Stab}\beta$ on X (i.e. the representation of $\operatorname{Stab}\beta$ in $\mathbb{C}^{n+1}$) without changing the action itself by multiplying it by this character (cf. [13], 8.11. This is equivalent to changing the momentum map from μ to $\mu+m\beta$). The weights of the representation then become $\alpha_0+m\beta, \ldots, \alpha_n+m\beta$.

Let $\{\Sigma_\gamma|\gamma\in\mathscr{C}\}$ be the stratification of X defined as in § 1 but with respect to this modified action of $\operatorname{Stab}\beta$. Note that $\mathscr{C}$ is the intersection of a positive

Weyl chamber for Stab β with the set of closest points to 0 in t of convex hulls of nonempty subsets of $\{\alpha_0+m\beta,\dots,\alpha_n+m\beta\}$.

Adapting the definitions of § 1 to the modified action of Stab β, let

$$Z_{\beta,\gamma}=\{(x_0;\dots;x_n)\in X \mid x_j=0 \text{ if } (\alpha_j+m\beta)\cdot\gamma\neq\|\gamma\|^2\}$$

and

$$Y_{\beta,\gamma}=\{x\in X \mid \lim_{t\to\infty}\exp(-it\gamma)\,x\in Z_{\beta,\gamma}\}.$$

Then a point $(x_0;\dots;x_n)\in X$ lies in $Y_{\beta,\gamma}$ if and only if

(2.17) (a) $x_j=0$ when $(\alpha_j+m\beta)\cdot\gamma<\|\gamma\|^2$, and

(b) there is some j_0 such that $(\alpha_j+m\beta)\cdot\gamma=\|\gamma\|^2$ and $x_{j_0}\neq 0$.

The stratum Σ_γ is defined by

$$\Sigma_\gamma=(\text{Stab }\beta)\,Y^{ss}_{\beta,\gamma}$$

where

$$Y^{ss}_{\beta,\gamma}=\{x\in X \mid \lim_{t\to\infty}\exp(-it\gamma)\,x\in Z^{ss}_{\beta,\gamma}\}$$

and $Z^{ss}_{\beta,\gamma}$ is the set of semistable points in $Z_{\beta,\gamma}$ under a suitable linearisation of the action of $(\text{Stab }\beta)\cap(\text{Stab }\gamma)$.

Let

$$\mathscr{D}=\{\gamma\in\mathscr{C} \mid \gamma\cdot\beta=(1+m)\,\|\beta\|^2\}.$$

When it is necessary to indicate that $\mathscr{D}$ depends on β it will be written as $\mathscr{D}(\beta)$.

(2.18) **Lemma.** *If m is sufficiently large then*

$$Y_\beta-Y^{ss}_\beta=\bigcup_{(1+m)\beta\neq\gamma\in\mathscr{D}}\Sigma_\gamma.$$

For the proof we need the following lemma whose proof is elementary.

(2.19) **Lemma.** *If m is sufficiently large then*

$$(\alpha_i+m\beta)\cdot\gamma<\|\gamma\|^2$$

whenever $\gamma\in\mathscr{C}$ and α_i is a weight such that

$$(\alpha_i+m\beta)\cdot\beta<\gamma\cdot\beta.$$

Moreover

$$(\alpha_i+m\beta)\cdot\gamma>\|\gamma\|^2$$

whenever

$$(\alpha_i+m\beta)\cdot\beta>\gamma\cdot\beta.$$

Thus if

$$(\alpha_i+m\beta)\cdot\gamma=\|\gamma\|^2$$

then

$$(\alpha_i+m\beta)\cdot\beta=\gamma\cdot\beta.$$

Proof of (2.18). It is easy to check that

$$Y_{\beta,(1+m)\beta} = Y_\beta$$

and

$$Y^{ss}_{\beta,(1+m)\beta} = Y^{ss}_\beta.$$

Since Y^{ss}_β is invariant under the action of Stab β this implies that

$$\Sigma_{(1+m)\beta} = (\text{Stab } \beta)\, Y^{ss}_\beta = Y^{ss}_\beta. \tag{2.20}$$

Thus it suffices to show that

$$Y_\beta = \bigcup_{\gamma \in \mathscr{D}} \Sigma_\gamma.$$

Suppose $\gamma \in \mathscr{D}$. Then it follows immediately from (2.17) and (2.19) that if m is large enough then

$$Y_{\beta,\gamma} \subseteq Y_\beta.$$

Hence

$$\Sigma_\gamma = (\text{Stab } \beta)\, Y^{ss}_{\beta,\gamma} \subseteq (\text{Stab } \beta)\, Y_\beta = Y_\beta.$$

Thus

$$Y_\beta \supseteq \bigcup_{\gamma \in \mathscr{D}} \Sigma_\gamma.$$

By [13], 12.16 applied to the stratification $\{\Sigma_\gamma | \gamma \in \mathscr{C}\}$ we have

$$Y_\beta = Y_{\beta,(1+m)\beta} \subseteq \bigcup_{\|\gamma\|^2 \geqq \|(1+m)\beta\|^2} \Sigma_\gamma.$$

If m is chosen large enough then any $\gamma \in \mathscr{C}$ satisfies

$$\|\gamma\|^2 \geqq \|(1+m)\beta\|^2$$

only if

$$\gamma \cdot \beta \geqq (1+m)\|\beta\|^2.$$

It is an immediate consequence of (2.19) that if

$$\gamma \cdot \beta > (1+m)\|\beta\|^2$$

then

$$Y_{\beta,\gamma} \cap Y_\beta = \phi$$

and hence

$$\Sigma_\gamma \cap Y_\beta \subseteq (\text{Stab } \beta)\, Y_{\beta,\gamma} \cap Y_\beta = \phi$$

because Y_β is invariant under Stab β. This shows that

$$Y_\beta \subseteq \bigcup_{\gamma \in \mathscr{D}} \Sigma_\gamma$$

and completes the proof of (2.18).

(2.21) **Corollary.** *Let W_β be any $K(\beta)$-invariant open neighbourhood of Y_β in X such that Y_β is closed in W_β, and let*

$$W_\beta^{ss} = W_\beta - \bigcup_{(1+m)\beta \neq \gamma \in \mathscr{D}} \Sigma_\gamma.$$

Then $W_\beta^{ss} \cap Y_\beta = Y_\beta^{ss}$. In particular W_β^{ss} is an open neighbourhood of Y_β^{ss} in X such that Y_β^{ss} is closed in W_β^{ss}.

Proof. This follows immediately from (2.18).

Let W_β and W_β^{ss} be as in (2.21). Just as in the first part of this section we can write

$$\mathscr{D} - \{(1+m)\beta\} = \{\gamma_1, \ldots, \gamma_t\}$$

so that

$$W_j = W_\beta - \bigcup_{j < i \leqq t} \Sigma_{\gamma_i}$$

is open in W_β when $0 \leqq j \leqq t$. Then by [8], (1.3) and (1.5) we get long exact sequences of equivariant intersection cohomology

$$\ldots \to IH^i_{K(\beta)}(W_j, W_{j-1}) \to IH^i_{K(\beta)}(W_j, W_j - Y_\beta) \to IH^i_{K(\beta)}(W_{j-1}, W_{j-1} - Y_\beta) \to \ldots$$

because

$$\begin{aligned} W_{j-1} \cup (W_j - Y_\beta) &= W_j - (Y_\beta \cap \Sigma_{\gamma_j}) \\ &= W_j - \Sigma_{\gamma_j} = W_{j-1}. \end{aligned}$$

(2.22) **Lemma.** *These long exact sequences split into short exact sequences*

$$0 \to IH^i_{K(\beta)}(W_j, W_{j-1}) \to IH^i_{K(\beta)}(W_j, W_j - Y_\beta) \to IH^i_{K(\beta)}(W_{j-1}, W_{j-1} - Y_\beta) \to 0$$

for each i and j.

Proof. It suffices to prove that the map

$$IH^i_{K(\beta)}(W_j, W_{j-1}) \to IH^i_{K(\beta)}(W_j, W_j - Y_\beta)$$

is injective for each i and j. Hence it suffices to prove that the composition of this map with the natural map

$$IH^i_{K(\beta)}(W_j, W_j - Y_\beta) \to IH^i_{K(\beta)}(W_j)$$

is injective. This composition fits into the long exact sequence

$$\ldots \to IH^i_{K(\beta)}(W_j, W_{j-1}) \to IH^i_{K(\beta)}(W_j) \to IH^i_{K(\beta)}(W_{j-1}) \to \ldots.$$

Since $W_{j-1} = W_j - \Sigma_{\gamma_j}$ where Σ_{γ_j} is closed in the open subset W_j of X and is a stratum of the stratification of X associated to the modified action of Stab β, the proof of (2.2) shows that this long exact sequence breaks up into short exact sequences

$$0 \to IH^i_{K(\beta)}(W_j, W_{j-1}) \to IH^i_{K(\beta)}(W_j) \to IH^i_{K(\beta)}(W_{j-1}) \to 0.$$

The result follows.

From this lemma we deduce that if $\beta = \beta_j$ then

$$IP_t^K(U_j, U_{j-1}) = t^{k(\beta)} IP_t^{K(\beta)}(W_\beta^{ss}, W_\beta^{ss} - Y_\beta^{ss}) \tag{2.23}$$
$$= IP_t^{K(\beta)}(W_\beta, W_\beta - Y_\beta) - \sum_{1 \leqq i \leqq t} IP_t^{K(\beta)}(W_i, W_{i-1}).$$

Here each $IP_t^{K(\beta)}(W_i, W_{i-1})$ has the same form as $IP_t^K(U_j, U_{j-1})$ and can be written as

$$IP_t^{K(\beta)}(W_i, W_{i-1}) = t^{k(\beta,\gamma)} IP_t^{K(\beta,\gamma)}(W_{\beta,\gamma}^{ss}, W_{\beta,\gamma}^{ss} - Y_{\beta,\gamma}^{ss})$$

where $\gamma = \gamma_i$ and $k(\beta,\gamma)$, $K(\beta,\gamma)$, $U_{\beta,\gamma} = W_{\beta,\gamma}^{ss}$ and $Y_{\beta,\gamma}^{ss}$ are defined in the same way as $k(\beta)$, $K(\beta)$, $U_\beta = W_\beta^{ss}$ and Y_β^{ss} but relative to the modified action of $\operatorname{Stab}\beta$ instead of the original action of G. Therefore (2.23) gives us an inductive formula for $IP_t^K(U_j, U_{j-1})$. This can be made into an explicit formula involving $IP_t^{K(\beta)}(W_\beta, W_\beta - Y_\beta)$ and other terms of the same form (see [13], § 5).

(2.24) **Lemma.**

$$IH^*_{K(\beta)}(W_\beta, W_\beta - Y_\beta) \cong IH^*(W_\beta, W_\beta - Y_\beta) \otimes H^*(BK(\beta))$$

where $BK(\beta)$ is the classifying space for $K(\beta)$.

Proof. Let T_0 be a maximal compact torus of $K(\beta)$. Then by [16], (2.16) there is a natural isomorphism

$$IH^*_{T_0}(W_\beta, W_\beta - Y_\beta) \cong IH^*(W_\beta, W_\beta - Y_\beta) \otimes H^*(BT_0).$$

This isomorphism respects the action of the Weyl group of $K(\beta)$ on both sides, and hence restricts to an isomorphism between the subspaces consisting of elements fixed by the Weyl group action. These subspaces are naturally isomorphic to

$$IH^*_{K(\beta)}(W_\beta, W_\beta - Y_\beta) \quad \text{and} \quad IH^*(W_\beta, W_\beta - Y_\beta) \otimes H^*(BK(\beta))$$

respectively. This proves the lemma.

Note that by [16], (2.9) and (2.10) we can write $IH^i(W_\beta, W_\beta - Y_\beta)$ as the hypercohomology group

$$\mathscr{H}^{-i}(Z_\beta, j_\beta^* i_\beta^! IC_X^\cdot)$$

of the complex of sheaves $j_\beta^* i_\beta^! IC_X^\cdot$ on Z_β, where $i_\beta\colon Y_\beta \to X$ and $j_\beta\colon Z_\beta \to Y_\beta$ are the inclusions and $IC_X^\cdot$ is the complex of intersection chains on X ([9], 2.1).

We have now proved the following theorem, which gives formulas for the intersection Betti numbers of the quotient $X//G$ when every semistable point of X is stable.

(2.25) **Theorem.** (a) *The equivariant intersection Poincaré series of X^{ss} satisfies*

$$IP_t^K(X^{ss}) = IP_t(X)\, P_t(BK) - \sum_{0 \neq \beta \in \mathscr{B}} t^{k(\beta)} IP_t^{K(\beta)}(U_\beta, U_\beta - Y_\beta^{ss})$$

where U_β is any $K(\beta)$-invariant neighbourhood of Y_β^{ss} in X such that Y_β^{ss} is closed in U_β.

(b) *The term* $IP_t^{K(\beta)}(U_\beta, U_\beta - Y_\beta^{ss})$ *in this expression is given by the inductive formula*

$$IP_t^{K(\beta)}(U_\beta, U_\beta - Y_\beta^{ss}) = IP_t(W_\beta, W_\beta - Y_\beta)\, P_t(BK(\beta)) - \sum_{(1+m)\beta \neq \gamma \in \mathscr{Q}(\beta)} t^{k(\beta,\gamma)}\, IP_t^{K(\beta,\gamma)}(U_{\beta,\gamma}, U_{\beta,\gamma} - Y_{\beta,\gamma}^{ss})$$

where W_β *is any* $K(\beta)$*-invariant open neighbourhood of* Y_β *in* X *such that* Y_β *is closed in* W_β.

(c) *The relative intersection Poincaré polynomial* $IP_t(W_\beta, W_\beta - Y_\beta)$ *depends only on the action of* K *near* Z_β. *More precisely*

$$IP_t(W_\beta, W_\beta - Y_\beta) = \sum_{i \geq 0} t^i\, \mathscr{H}^{-i}(Z_\beta, j_\beta^* i_\beta^! IC_X^\cdot).$$

(d) *Finally if every semistable point of* X *is stable then*

$$IP_t(X//G) = IP_t^K(X^{ss}).$$

(2.26) *Remark.* We know that the intersection of Z_β with any nonsingular quasi-projective subvariety of X is nonsingular: in particular this is true of the strata of a Whitney stratification of X. However Z_β need not meet these strata transversely. If this does happen and the inclusion of Z_β in X is normally nonsingular for every β then

$$IH^*_{\mathrm{Stab}\,\beta}(U_\beta, U_\beta - Y_\beta^{ss}) \cong IH^{*-2\lambda(\beta)}_{\mathrm{Stab}\,\beta}(Z_\beta^{ss})$$

where $\lambda(\beta)$ is the complex codimension of Y_β^{ss} in X ([9], 5.4.1) so that (2.11) gives exactly the same inductive formula for $IP_t^G(X^{ss})$ as one has in the nonsingular case (see [13], 8.12) except that ordinary cohomology is replaced by intersection cohomology throughout. In particular if the singular points of X are all semistable then this is true.

(2.27) *Example.* Let X be the cubic hypersurface in $\mathbb{P}_4$ defined by

$$x^3 + y^3 + z^3 = u^2 v.$$

Let $G = \mathbb{C}^*$ act on X via

$$t \to \mathrm{diag}(t, t, t, t^4, t^{-5}).$$

Let

$$IP_t(X) = 1 + a_1 t + a_2 t + a_3 t^3 + a_2 t^4 + a_1 t^5 + t^6.$$

Of course $a_1 = 0$ and $a_2 = 1$ by the Lefschetz hyperplane theorem ([9], 5.3) but we will be able to deduce this and the value of a_3 later.

As G has rank 1 the stratification $\{S_\beta | \beta \in \mathscr{B}\}$ of X is indexed by 0 together with the set of weights of the representation of G, i.e. by 0, 1, 4 and -5 if we identify Lie S^1 with $\mathbb{R}$ in the usual way. We have

$$S_1 = Y_1 = \{(x:y:z:u:0) \in \mathbb{P}_n | (x,y,z) \neq 0,\ x^3 + y^3 + z^3 = 0\},$$
$$Z_1 = Z_1^{ss} = \{(x:y:z:0:0) \in \mathbb{P}_n | x^3 + y^3 + z^3 = 0\},$$
$$S_4 = Y_4 = Z_4 = Z_4^{ss} = \{(0:0:0:1:0)\}$$

and

$$S_{-5}=Y_{-5}=Z_{-5}=Z^{ss}_{-5}=\{(0:0:0:0:1)\}.$$

Note that G acts with finite stabilisers on $S_0=X^{ss}$, so that $X^{ss}=X^s$. Hence Theorem (2.25) can be applied to calculate $IP_t(X//G)$. One finds that

$$\begin{aligned}IP_t(X//G)&=(1+a_1t+a_2t^2+a_3t^3+a_2t^4+a_1t^5+t^6)(1-t^2)^{-1}\\&\quad-t^2(1+2t+t^2)(1-t^2)^{-1}-t^6(1-t^2)^{-1}-t^6(1-t^2)^{-1}\\&=(1+a_1t+(a_2-1)t^2+(a_3-2)t^3+(a_2-1)t^4+a_1t^5-t^6)(1-t^2)^{-1}.\end{aligned}$$

Since $IP_t(X//G)$ must be a polynomial of degree 4 in t satisfying Poincaré duality ([9], 7.1) and a_1, a_2 and a_3 are non-negative integers we deduce that $a_1=0$, $a_2=1$ and $a_3=2$, so that

$$IP_t(X//G)=(1-t^6)(1-t^2)^{-1}=1+t^2+t^4.$$

This example shows in a very trivial way how it is possible to deduce information about the intersection homology of both X and $X//G$ from (2.25) simply using the fact that $IP_t(X//G)$ is a symmetric polynomial of degree the dimension of $X//G$.

(2.28) *Remark.* Finally let us consider briefly the more general case when we only assume that X^s is nonempty and allow X to have semistable points which are not stable.

There exists a canonically defined projective variety $\tilde{X}$ with a linear action of G and a birational G-morphism $\pi\colon \tilde{X}\to X$ such that $\tilde{X}^{ss}=\tilde{X}^s$. This variety $\tilde{X}$ is constructed in [14] in the case when X is nonsingular, and the construction extends easily to the singular case. There is an induced birational morphism $\pi_G\colon \tilde{X}//G\to X//G$ of quotient varieties. When X is nonsingular the quotient $\tilde{X}//G$ may be regarded as a partial desingularisation of $X//G$ since $\tilde{X}//G$ has only finite quotient singularities whereas the singularities of $X//G$ are in general much worse.

The decomposition theorem of Beilinson, Bernstein, Deligne and Gabber ([2], 6.2.5) tells us that the intersection cohomology $IH^*(X//G)$ of $X//G$ is a direct summand of $IH^*(\tilde{X}//G)$. One has an explicit description of the birational morphism $\pi_G\colon \tilde{X}//G\to X//G$ as the composition of a sequence $\phi_1,\dots,\phi_r$ of blow-ups whose centres are quotients of the same form as $\tilde{X}//G$ (see [14]), and the decomposition theorem implies that

$$IH^*(\tilde{X}//G)\cong\Big(\bigoplus_{1\leqq j\leqq r} IH^*(\tilde{\mathcal{N}}_j)/IH^*(\mathcal{N}_j)\Big)\oplus IH^*(X//G) \tag{2.29}$$

where $\mathcal{N}_j$ is any open neighbourhood of the centre of ϕ_j and $\tilde{\mathcal{N}}_j$ is its inverse image under ϕ_j (cf. the proof of [15], 2.8).

When X is nonsingular the centre of each blow-up ϕ_j has a neighbourhood $\mathcal{N}_j$ such that $\mathcal{N}_j$ and $\tilde{\mathcal{N}}_j$ are topologically fibre bundles over the centre. The fibres are, respectively, the affine cone over a quotient of the same form as $X//G$ and the blow-up of this cone at the vertex. These facts, together with (2.29), are used in [15] to obtain an inductive formula for the intersection Betti

numbers of $X//G$ which involves the Betti numbers of $\tilde{X}//G$. The latter can be computed using 1.2 because $\tilde{X}^{ss}=\tilde{X}^{s}$ (see also [14]).

When X is singular one can hope first to use Theorem (2.25) to compute the intersection Betti numbers of $\tilde{X}//G$, and then to apply (2.29) to obtain the intersection Betti numbers of the original quotient $X//G$. Unfortunately in general it is very hard to work out the contributions of the terms $IH^*(\tilde{\mathcal{N}}_j)/IH^*(\mathcal{N}_j)$ which appear in (2.29). In the special case when the centres of all the blow-ups in the construction of $\tilde{X}^{ss}$ from X^{ss} are transverse to the singularities of X, then the formulas of [14] and [15] are valid if cohomology is replaced by intersection cohomology throughout.

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Oblatum 5-XI-1986